

Project Briefing

Proxy Validation for Conversational Multi-Agent Systems Built from Small Language Models

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How to use this document

I am giving this to an AI assistant as background before asking it questions about my research project. Please treat everything below as established context about work already underway, not as something to independently verify or second-guess. The three things I most need explained clearly, if you are helping me talk this through with someone else, are: the three multi-agent topologies I am testing, why those three specifically and not others, and exactly what kind of power measurement the project requires on each platform. Sections 3, 4, and 5 cover those in that order.

1 Background: The Review Paper

Before this project existed, I co-authored a review paper, *Reassessing Agentic Optimization: A Review of Proxies in Multi-Agent Systems*, with Ashwath Rao B. This project is the direct empirical follow-up to that review, so understanding what the review found is necessary to understand why this project is shaped the way it is.

1.1 What the review found

Efficiency in multi-agent large language model (LLM) systems is almost universally reported using proxies: token count, wall-clock time, and monetary cost. Real, hardware-measured energy is reported far less often. The review organised the existing literature along three axes:

- **Granularity** — whether the work concerns a single model responding to a single request, or a coordinated group of agents.
- **Lifecycle stage** — whether the intervention happens before deployment (training, fine-tuning) or at inference time (decoding configuration, prompt design, agent orchestration).
- **Measurement type** — whether efficiency is reported as a proxy (tokens, time, cost, FLOPs, model size), as an estimate (derived from board power limits rather than per-process sampling), or as real, directly measured energy from a hardware counter or external power meter.

1.2 The specific gap

Classifying the literature this way exposed a structural hole. Of the multi-agent systems reviewed that report on efficiency, the majority report only proxies and never measure real energy at all. Of the small number of multi-agent studies that do measure real energy, every one of them is a coding, tool-using, or local-device system. None of them is a conversational system: none use debate between peer agents, none use a solver being checked by a critic, and none report a hierarchical planner delegating to workers.

This matters because conversational multi-agent systems are not a minor category. They make up a large share of the current agent-efficiency literature. The review's central finding is that this specific category, conversational MAS, is a cell in the taxonomy with no published direct energy measurement at all. This project exists to fill that cell.

1.3 Why the gap is not a minor omission

The review also surveyed what happens when proxies *have* been checked against real energy, mostly in single-model settings, because that comparison is the reason the gap cannot be assumed harmless. Token count, the most commonly reported metric in agent-efficiency work, explained only a moderate share of measured energy variance (R^2 roughly 0.41–0.52) in a direct comparison against real hardware measurement, while wall-clock time explained substantially more (R^2 above 0.91), despite being the less commonly reported of the two. Separately, in multi-agent coding systems where real energy has been measured, token and pricing proxies were found to systematically underestimate real GPU energy, because output tokens carry disproportionately more energy cost than input tokens do. Compute-based proxies (FLOPs) and compression techniques (quantization) have also failed to track real energy consistently: quantization has been observed to *increase* energy use in some cases because it reduces throughput rather than genuinely reducing computation.

The review’s position is not that proxies are known to fail in conversational MAS. It is that proxies have failed often enough elsewhere that using them in conversational MAS without checking is an unverified assumption, and that assumption has never been tested because no one has directly measured energy in that setting.

1.4 The review’s stated future directions

The review closed by naming four concrete follow-up questions. This project answers three of them directly:

1. Whether the output-token energy asymmetry observed in multi-agent *coding* systems also holds in natural-language conversational exchanges, where the content is reasoning rather than code or tool calls.
2. Whether sampling temperature influences energy as a genuine *cause*, plausibly through its effect on retry frequency, rather than merely correlating with it, as prior work has only shown a correlation without connecting the mechanism.
3. Whether monetary cost, which is the entire optimisation target of at least one existing multi-agent efficiency system, has ever actually been validated against measured energy in multi-turn agentic workloads. It has not, and this project does so.

The fourth direction the review raises, break-even accounting for whether an optimisation’s energy savings exceed the energy spent finding it, is out of scope for this project and left for future work.

2 The Project: Overview

This project performs direct, hardware-based energy measurement on conversational multi-agent systems built from small language models (SLMs), and checks whether token count, wall-clock time, and monetary cost actually track that measured energy. It does not assume the proxies fail; it measures and reports what the relationship actually is.

2.1 Why SLMs specifically

The project restricts itself to small language models because they are the class of model this specific question can be answered on with real, physically instrumented hardware rather than assumed or software-only proxies. A small model can be run on a fully owned edge device, which makes it possible to place an external, independently calibrated power sensor directly on the device’s power input. That is not possible on a shared, remotely accessed datacentre GPU, which is why the project design uses two platforms with two different measurement scopes rather than one.

2.2 Platforms

- **Jetson Orin NX** (primary). Fully owned hardware. Carries an external, physically installed power measurement rig in addition to its own onboard power reporting.
- **NVIDIA A100** (secondary). Reached over shared SSH access, without root privileges and without exclusive use of the device. No physical modification is possible. Energy is read from the GPU’s own built-in hardware counter.

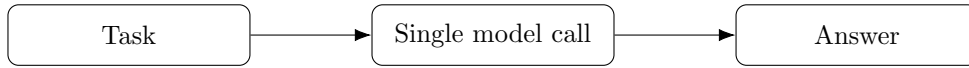
2.3 Design

The experiment is a full factorial over four conditions (a single-call baseline plus the three multi-agent topologies described in Section 3) crossed with three sampling temperatures (0.2, 0.7, 1.0), giving twelve configurations. The full design runs on the Jetson NX first, since it is the fully controlled, primary dataset. The same twelve configurations are then attempted on the A100 as a secondary check, with any call found to have run alongside another user’s job on the shared GPU flagged and excluded rather than silently kept. The project runs on a six-week timeline.

3 The Three Topologies

A “topology” here means a fixed pattern for how multiple agent calls are structured, sequenced, and combined to answer one task. All three below are *conversational*: the agents exchange natural-language reasoning with each other, not tool calls or code execution. That is the specific property the review found unmeasured, and it is why these are not interchangeable with the coding or tool-using multi-agent systems that already have energy measurements in the literature.

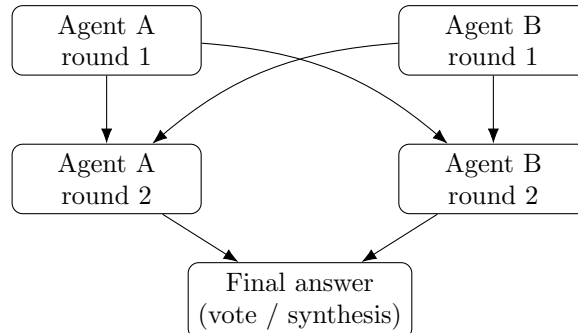
Every topology is also run against a **single-call baseline**: one model, one call, no coordination at all. The baseline is not a topology in the same sense as the other three; it is the zero-coordination reference point every topology is measured against.



Baseline: no coordination, no peer exposure, no revision.

3.1 Debate

Two or more agent instances independently answer the same task in parallel, with no communication in the first round. In each subsequent round, every agent is shown the other agents’ answers from the previous round and is asked to reconsider its own answer in light of them, then produces a revised answer. This repeats for a fixed number of rounds. The final answer is taken either by majority vote across the agents’ final-round answers or by a separate synthesis step. All roles are symmetric: no agent has authority over another, and every agent both answers and critiques.

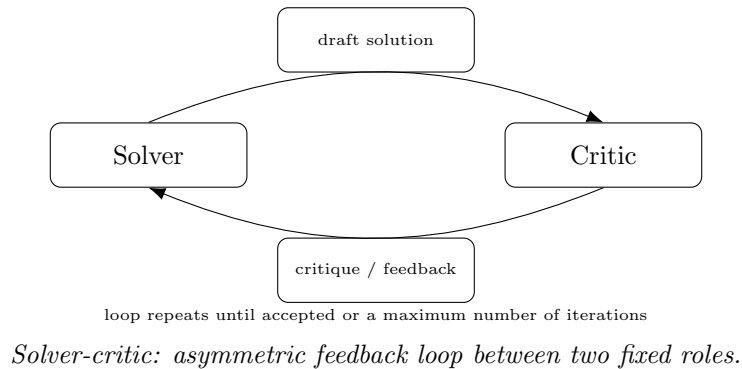


Debate: symmetric peers, each round conditioned on the other agent’s prior answer.

Termination: a fixed number of rounds (not a variable stopping condition), which keeps the topology’s structure — and therefore its call count — stable across items and comparable across configurations.

3.2 Solver-Critic

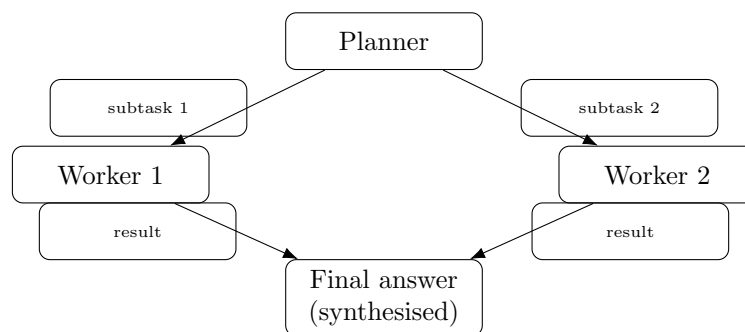
Two asymmetric, fixed roles. The *solver* produces an attempt at the task. The *critic*, a separate agent call, evaluates that attempt: it checks correctness, flags specific errors, and returns feedback. The solver revises its answer using that feedback and resubmits it to the critic. This loop repeats until the critic accepts the solution or a maximum number of iterations is reached, at which point the last solver answer is taken as final.



Termination: a stopping condition, either the critic’s acceptance or an iteration cap. Unlike debate’s fixed round count, this topology’s call count can genuinely vary between items, which is itself relevant to the project: it is exactly the kind of retry mechanism the review’s future-direction on temperature-as-cause is asking about.

3.3 Planner-Worker

A hierarchical, asymmetric pattern. A *planner* agent decomposes the incoming task into two or more subtasks. Each subtask is dispatched to a separate *worker* agent call, which executes only its assigned subtask and returns a result. The planner collects every worker’s result and synthesises them into one final answer. Workers do not communicate with each other, only with the planner.



Planner-worker: one coordinating agent, subtasks delegated to independent workers.

Termination: once every dispatched subtask has returned a result and the planner has synthesised them; the call count is determined by how many subtasks the planner decides to create, which can itself vary by item.

4 Why These Specific Topologies

These three were not chosen arbitrarily, and they were not chosen for variety’s own sake. Two considerations drove the choice.

4.1 They are named in the gap the review identified

Debate and solver-critic are named explicitly, by name, as the two subtypes of conversational multi-agent system for which the review found no energy-measurement study at all. Planner-worker delegation is the third major coordination pattern the review discusses as a granularity-defining structure in the agent-efficiency literature, and it is likewise absent from the small set of studies that do measure real multi-agent energy, which are confined to coding and tool-using systems. Testing these three specific topologies is therefore not a convenient sample of possible MAS structures; it is a direct, targeted attempt to fill the exact hole the review identified, rather than a nearby but different hole.

4.2 They are structurally different from each other on purpose

The three topologies differ along the axis that actually matters for whether a proxy-validity finding will generalise: how coordination is structured.

- **Debate** is *symmetric*: every agent has the same role and the same authority.
- **Solver-critic** is *asymmetric and iterative*: two fixed, unequal roles in a feedback loop with a variable number of iterations.
- **Planner-worker** is *hierarchical*: one coordinating role and multiple subordinate roles that never speak to each other directly.

If a proxy such as token count turns out to explain measured energy well, or poorly, consistently across all three of these structurally distinct patterns, that is a far stronger and more general claim than a finding that only held for one topology, or for three topologies that were really variations on the same coordination idea. Conversely, if a proxy’s validity *changes* depending on which of these three structures is used, that is itself an important, specific, and previously unreported finding, since the review’s account of why single-model results might not transfer to multi-agent settings is built on exactly these kinds of structural differences: retries, re-derivation, and coordination overhead that a single model call never has to pay for.

4.3 The baseline’s role

The single-call baseline is not being compared against the other three to ask “which topology is best.” It exists to isolate how much of any observed proxy-energy relationship is attributable to multi-agent coordination itself, as opposed to being a property of the underlying model that would show up even with no coordination at all. Without the baseline, there would be no way to tell whether a proxy’s failure (or success) in a multi-agent topology is caused by the topology, or was already present in the model being called.

5 Power Measurement Requirements

5.1 Why proxies are not sufficient here

The entire premise of this project is that tokens, wall-clock time, and monetary cost are proxies for energy, not energy itself, and the review’s own findings show those proxies cannot be assumed to track energy reliably. Answering the project’s questions requires a real, physically measured energy value for every single model call, against which the three proxies can be checked. Nothing

software-reported alone is sufficient as the reference; it has to be checked against, or in one case is, actual hardware measurement.

5.2 The critical distinction: whole-device energy versus GPU-only energy

The two platforms in this project do not measure the same physical quantity, and this has to be understood clearly before comparing anything between them.

- On the **Jetson NX**, the external hardware sensor sits on the board’s DC power input, upstream of everything on the board. It therefore captures *whole-device* energy: the system-on-chip (which includes the GPU, CPU cores, and other accelerators together, since this platform has no separate discrete GPU), memory, the carrier board’s own regulators, storage, and the fan. There is no way to isolate “just the GPU’s share” from this reading alone.
- On the **A100**, the GPU’s own built-in hardware energy counter is scoped to the physical GPU package only. It does not include the host server’s CPU, system memory, storage, networking, or the server’s power-supply losses. This is *GPU-only* energy, with the rest of the machine entirely excluded.

This means a raw energy number from the NX and a raw energy number from the A100 are not directly comparable as absolute quantities; the NX number is structurally larger because it includes components the A100 number never captures. Any comparison between the two platforms needs to be framed around *whether the proxies track energy similarly on each platform*, not around comparing the absolute energy values to each other.

5.3 Jetson NX: two measurement paths

- **Hardware path (ground truth).** An external, independently calibrated current sensor is placed in series with the board’s power input and sampled at roughly 1–1.5 kHz by an isolated microcontroller. This produces the whole-device energy reading described above, and because it is calibrated against a reference meter rather than trusted as-is, it serves as the ground truth the rest of the project’s energy claims are checked against.
- **Software path (onboard).** The platform’s own internal power-rail monitors report energy for specific internal domains (roughly: the GPU/SoC compute domain, the CPU domain, and the main system rail), sampled far less frequently than the hardware path and covering a narrower scope than the whole device.

Both paths are needed together, not instead of each other: comparing them against each other is how the project checks whether the platform’s own self-reported energy number, the kind every other paper in this space relies on when they say “we measured energy,” can actually be trusted.

5.4 A100: one measurement path, under real constraints

Only the GPU’s own hardware energy counter is available, since there is no physical or port-level access to the machine to attach external instrumentation, and access is over shared SSH without administrative privileges. Two consequences follow directly from that: clock speed cannot be locked or fixed the way it can on the fully owned NX, so it is logged as a variable instead and accounted for statistically rather than controlled experimentally; and because the GPU may not be exclusively allocated, each call has to be checked, both immediately before and immediately after it runs, for whether another user’s job was also resident on the device. Any call where that check fails is excluded from the primary results rather than kept and treated as clean.

5.5 Attributing energy to a single call

An energy reading is only useful if it can be tied to one specific model call, not to a whole session or a whole block of calls. On the NX, this is done with a hardware trigger line: a signal is set high the instant a call is dispatched and set low the instant it returns, read directly by the same sampler that is recording the current, so the energy reading and the call boundary share a single clock and cannot drift apart from each other. On the A100, the hardware energy counter is read immediately before dispatching a call and again immediately after it returns, and the difference between the two readings is that call's energy.

5.6 What still needs to be controlled

Beyond the raw measurement, the reported energy for each call also needs the surrounding device state accounted for: idle power is measured before and after each block of calls so that background power draw can be subtracted from the reading, and on the NX, device temperature is checked before each call and logged, since a device running hot draws more power for the same work than one at its normal operating temperature.